

Vincent Harkins

vharkins1.github.io | linkedin.com/in/vincent-harkins | leetcode.com/vincentharkins

EDUCATION

University of California, Santa Cruz

Master of Science (M.S.) in Computer Science and Engineering

Santa Cruz, CA

Exp. Dec 2026

University of California, Santa Cruz

Bachelor of Science (B.S.) in Computer Engineering

Santa Cruz, CA

Sep 2021 – June 2025

- **Relevant Coursework:** Graduate Analysis of Algorithms (CSE 201), Programming Abstractions, Computer Systems and C Programming, Principles of Computer Systems Design, Computer Architecture, Intro to Computer Networks, Network Programming, Capstone Project I & II

TECHNICAL SKILLS

Languages: C, C++, Python, Swift, JavaScript, Java, C#, TypeScript, Bash

Frameworks & Tools: Node.js, React, **snnTorch**, PyTorch, Supabase, Ollama and vLLM, Vivado, PostgreSQL, Git, VSCode

EXPERIENCE

Neuromorphic Computing Group

Research Assistant

Santa Cruz, CA

Sep 2025 – Present

- Conducting FFN-Attention disaggregation research within the **vLLM** library for **Qernel AI** to analyze and optimize neural network architectures.
- Optimized Leaky Integrate-and-Fire neuron computation in **snnTorch** by replacing standard cumulative sums with convolution-based operations, achieving significant speedup in exchange for minimal information loss.
- Contributing to GPU energy profiling to analyze and optimize power usage during AI model inference.

Learning Support Services (LSS) at UCSC

Tutor (Logic Design / CSE 100)

Santa Cruz, CA

Jan 2025 – June 2025

- Tutored approximately 6 students in logic design (CSE 100), holding 3 sessions per week.
- Designed supplementary course material to build student confidence and improve performance with complex course content.

PROJECTS

FireForm (UN Reboot the Earth Hackathon - 1st Place) | React, OpenAI Whisper, Ollama

Nov 2025

- Won 1st place developing a wildfire detection platform; awarded 6-month coaching with **Salesforce**.
- Utilized Whisper (via OpenAI) for input processing and **Ollama** to serve Llama 3.2 7b for local inference.
- Built a responsive frontend with React to facilitate real-time user interaction.

SlugCruise (Ride Share App) | React, Node, Vite, Vercel, Supabase

Jan 2026

- Built the frontend using React, Node, and Vite (hosted on Vercel) for a campus-centric ride-share platform.
- Leveraged **Supabase** to handle the PostgreSQL database and secure student authentication.
- Designed real-time geolocation features to allow students to offer and request rides safely.

Angry Uncle Bot | React, Vite, OpenAI API

June 2025 – Present

- Developed a conversational web tool using OpenAI's ChatGPT API with Vercel serverless deployment.
- Engineered prompt instructions to ensure tone consistency and accuracy in responses.

Smart Pill Dispenser | React Native, Firebase, Embedded Systems

Jan 2025 – June 2025

- Led the development of a mobile-friendly web app for a smart pill dispenser system.
- Designed Firestore database schema and coordinated integration with embedded hardware.